MILITARY SPECIFICATION SHEET

TRANSFORMER, POWER, STEP-DOWN,
THREE PHASE, 50 Hz, TF5S03ZZ, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the transformer described herein shall consist of this document and the latest issue of Specification MIL-T-27.

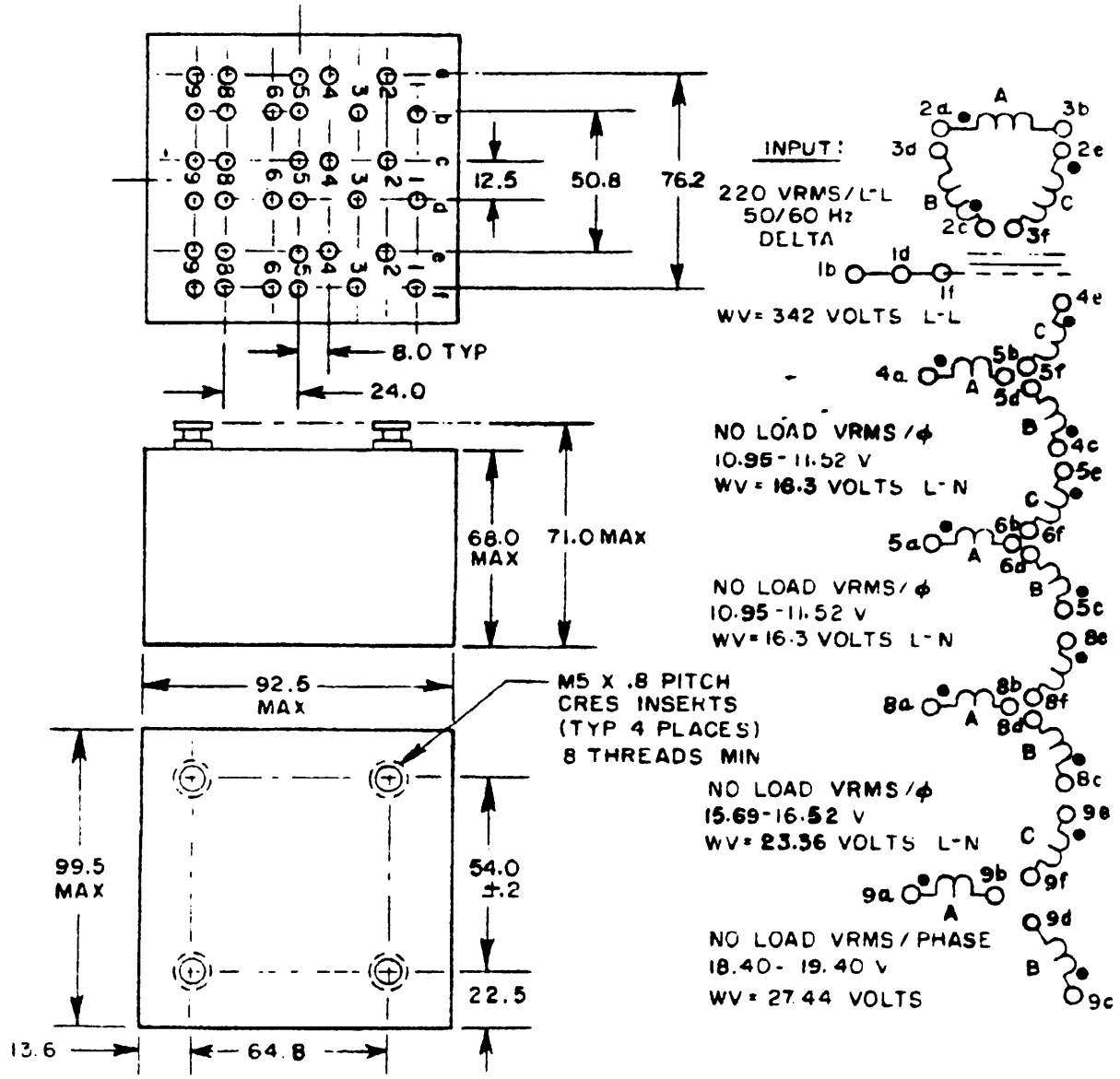


FIGURE 1. Dimensions and configuration.

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At reduced barometric pressure: 300 V rms.

MM	INCHES
.2	.008
.8	.031
8.0	.312
12.5	.488
13.6	.530
22.5	.878
24.0	.936
50.8	1.98
54.0	2.11
64.8	2.53
68.0	2.65
71.0	2.77
76.2	2.97
92.5	3.61
99.5	3.88

NOTES:
1. Dimensions are in millimeters.
2. English equivalents are given for general information only and are based upon 1.00 mm = .039 inches.
3. Tolerance on all dimensions shall be +0.50 mm(.02 inch) unless otherwise specified.
4. Terminals: 33 turret type, each shall be capable of accepting a minimum of one #22 AWG wire.
5. Mounting: A minimum bolt entry shall engage at least 8 metal threads. A minimum bolt entry shall avoid contact with the case and/or potting material.
6. Inserts shall have a minimum of 8 metal threads.

FIGURE 1. Dimensions and configuration - Continued.

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REQUIREMENTS: (When numbers in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding).

Electric ratings:
Primary voltage:
(2a-3b) (2e-3f) (2c-3d): 220 Vrms ±10 percent, line to line, 3 phase delta connected, 50 Hz ±5 percent.
Secondary voltage:
(4a-5b) (4e-5f) (4c-5d): 10.95 - 11.52 volts, 41 mA rms line to neutral wye connected.
(5a-6b) (5e-6f) (5c-6d): 10.95 - 11.52 volts, 204 mA rms line to neutral wye connected.
(8a-8b) (8e-8f) (8c-8d): 15.69 - 16.52 volts, 163 mA rms line to neutral wye connected.
(9a-9b): 18.4 - 19.4 volts, 300 mA rms.
(9c-9d): 18.4 - 19.4 volts, 70 mA rms.
(9e-9f): 18.4 - 19.4 volts, 6.8 mA rms.

Working voltage (peak):
(2a-3b) (2e-3f) (2c-3d): 342 volts line to line.
(4a-5b) (4e-5f) (4c-5d) (5a-6b) (5e-6f) (5c-6b): 16.3 volts line to neutral.
(8a-8b) (8e-8f) (8c-8d): 23.36 volts line to neutral.
(9a-9b) (9c-9d) (9e-9f): 27.44 volts.

Design and construction:
Dimension and configuration: See figure 1.
Duty cycle: Continuous.
Case: Metal base and epoxy, encapsulated.
Terminals: Thirty-three solderable turret type capable of accepting a minimum of one No. 22 AWG wire.
Height: 3.0 mm, maximum.
Weight: 5 pounds, maximum.
Operating temperature range: -55° to +130°C.
Altitude: 10,000 feet, maximum.

Terminal strength: MIL-STD-202, method 211, test condition A, 5 pounds.

Dielectric withstanding voltage (at sea level):
Primary: 1,500 V rms.
Secondary: 500 V rms.

Electrical characteristics:
No load: With 220 V rms, (2a-3b) (2c-3d) (2e-3f) line to line, 3 phase delta connected, 50 Hz.
Current in (2a-3b) (2c-3d) (2e-3f) line to line: 0.035 ampere rms each phase, maximum.
Power in (2a-3b) (2e-3d) (2e-3f) line to line: 8.42 watts each phase, maximum.
Voltage across (4a-5b) (4c-5d) (4e-5f) (5a-6b) (5c-6d) (5e-6f) wye connected:
11.52 volts line to neutral maximum.
(8a-8b) (8c-8d) (8e-8f) wye connected 16.5 volts line to neutral maximum.
(9a-9b) (9c-9d) (9e-9f): 19.4 volts, maximum.

Rated load: With 220 Vrms, line to line, 3-phase delta connected, 60 Hz across
(2a-3b) (2c-3d) (2e-3f), the voltage across:
(4a-5b) (4c-5d) (4e-5f) wye connected shall be 10.95 volts, minimum, at 41 mA line to neutral.
(5a-6b) (5c-6d) (5e-6f) wye connected shall be 10.95 Vrms, minimum, at 204 mA line to neutral.
(8a-8b) (8c-8d) (8e-8f) wye connected shall be 15.69 volts, minimum, at 163 mA line to neutral.
(9a-9b) shall be 18.4 V, minimum, at 300 mA.
(9c-9d) shall be 18.4 volts, minimum at 70 mA.
(9e-9f) shall be 18.4 volts, minimum at 6.8 mA.
DC resistance:
(2a-3b) (2c-3d) (2e-3f): 125 ohm, maximum line to line.
(4a-5b) (4c-5d) (4e-5f): 2.9 ohm, maximum line to neutral.
(5a-6b) (5c-6d) (5e-6f): 3.0 ohm, maximum line to neutral.
(8a-8b) (8c-8d) (8e-8f): 4.4 ohm, maximum line to neutral.
(9a-9b) (9c-9d) (9e-9f): 4.4 ohm, maximum.

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Temperature rise: 50°C with 264 Vrms, line to line, 3-phase delta connected, 60 Hz across (2a-3b) (2c-3d) (2e-3f) at an ambient temperature of 80°C.

Salt spray: MIL-STD-202, method 101, test condition B.

Vibration, high frequency: MIL-STD-202, method 204.

Shock (specified pulse): MIL-STD-202, method 213, test condition K.

Marking location: See figure 1.

Part number: D27/272-01.

Substitution data (information only): This military part is substitutable for FSCM 55196, part number 11215. This information in no way implies that the manufacturer's part is substitutable for the military part. This information is not intended to be complete.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR
Navy - SH, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC, AS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5950-0561-10)

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OMB Approval No 22-R255

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DOCUMENT IDENTIFIER AND TITLE: DOD-T-27/272 -TRANSFORMER, POWER, STEP-DOWN, THREE PHASE, 50 Hz, TF50322, METRIC

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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